United States Patent Application Publication	20250284454
Kind Code	A1
Publication Date	September 11, 2025
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CONTROL SYSTEM AND HEAD-MOUNTED DISPLAY

Abstract

A control system includes: a head-mounted display that is worn by a user riding in a vehicle, and provides the user with at least a display image corresponding to an image acquired by a vehicle exterior camera that is provided in the vehicle and captures an image of the periphery of the vehicle; a detection unit that detects a gesture performed by the user, based on an image acquired by a vehicle interior camera that is provided in the vehicle and captures an image of the interior of a vehicle cabin; and a volume control unit that executes control of adjusting the volume of a speaker provided in the head-mounted display based on the gesture detected by the detection unit.

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Family ID: **96931659**

Appl. No.: **19/063957**

Filed: **February 26, 2025**

Foreign Application Priority Data

JP	2024-034094	Mar. 06, 2024

Publication Classification

Int. Cl.: **G06F3/16** (20060101); **G02B27/01** (20060101); **G06F3/01** (20060101)

U.S. Cl.:

CPC **G06F3/165** (20130101); **G02B27/0172** (20130101); **G06F3/017** (20130101); G02B2027/0138 (20130101)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is based upon and claims the benefit of priority from Japanese Patent Application No. 2024-034094 filed on Mar. 6, 2024, the contents of which are incorporated herein by reference.

BACKGROUND OF THE INVENTION

Field of the Invention

[0002] The present disclosure relates to a control system and a head-mounted display.

Description of the Related Art

[0003] JP 7243639 B2 discloses an information processing apparatus in which a headphone and a smartphone are capable of wirelessly communicating with each other. In the information processing apparatus, the headphone includes a motion sensor capable of detecting the behavior of a user. The smartphone can be operated in response to the behavior of the user detected by the motion sensor. The smartphone can provide a game, a video, and the like to the user.

SUMMARY OF THE INVENTION

[0004] JP 7243639 B2 merely discloses that the smartphone is operated in response to the behavior of the user detected by the motion sensor provided in the headphone. In recent years, there has been a demand for technologies that enable a head-mounted display to be used satisfactorily in a vehicle cabin.

[0005] The present invention has the object of solving the aforementioned problem.

[0006] A control system according to a first aspect of the present disclosure comprises: a head-mounted display that is worn by a user riding in a vehicle, the head-mounted display being configured to provide the user with at least a display image corresponding to an image acquired by a vehicle exterior camera provided in the vehicle and configured to capture an image of a periphery of the vehicle; a detection unit configured to detect a gesture performed by the user, based on an image acquired by a vehicle interior camera provided in the vehicle and configured to capture an image of an interior of a vehicle cabin; and a volume control unit configured to execute control of adjusting a volume of a speaker provided in the head-mounted display, based on the gesture detected by the detection unit.

[0007] A head-mounted display according to a second aspect of the present disclosure is worn by a user riding in a vehicle, the head-mounted display comprising: a display unit configured to provide the user with at least a display image corresponding to an image acquired by a vehicle exterior camera provided in the vehicle and configured to capture an image of a periphery of the vehicle; a speaker configured to provide the user with a sound; a detection unit configured to detect a gesture performed by the user, based on an image acquired by a vehicle interior camera provided in the vehicle and configured to capture an image of an interior of a vehicle cabin; and a volume control unit configured to execute control of adjusting a volume of the speaker based on the gesture detected by the detection unit.

[0008] According to the present invention, the head-mounted display can be used satisfactorily in the vehicle cabin.

[0009] The above and other objects, features, and advantages of the present invention will become more apparent from the following description when taken in conjunction with the accompanying drawings, in which a preferred embodiment of the present invention is shown by way of illustrative example.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] FIG. **1** is a schematic configuration diagram of a control system according to an embodiment of the present invention;

[0011] FIG. **2** is a functional block diagram of a head-mounted display;

[0012] FIG. **3** is a flowchart of a volume adjustment process;

[0013] FIG. **4**A is a diagram showing a user performing a first gesture;

[0014] FIG. **4**B is a diagram showing the user performing a second gesture;

[0015] FIG. **5** is a diagram showing the user performing a third gesture;

[0016] FIG. **6**A is a diagram showing the user performing the first gesture with both hands;

[0017] FIG. **6**B is a diagram showing the user performing the second gesture with both hands;

[0018] FIG. **7** is a diagram showing the user performing the first gesture with one hand and lowering the other hand;

[0019] FIG. **8** is a diagram showing the user performing the first gesture with one hand and pointing in a direction with the other hand; and

[0020] FIG. **9** is a diagram showing the user performing the third gesture with one hand and pointing in a direction with the other hand.

DETAILED DESCRIPTION OF THE INVENTION

[0021] It is desirable that a head-mounted display allows a user to be immersed in content such as a game or a video.

[0022] Both eyes of the user wearing the head-mounted display are covered with the display surface of the head-mounted display. Therefore, the user wearing the head-mounted display in the vehicle cabin cannot see the scenery outside the vehicle. When the vehicle travels in this state, the user may feel motion sickness.

[0023] The embodiment described herein makes it less likely that the sense of immersion of the user in content such as a game or a video will be impaired. In addition, the embodiment described herein suppresses motion sickness of the user wearing a head-mounted display in a vehicle cabin.

1. Configuration of Control System **12**

[0024] FIG. **1** is a schematic configuration diagram of a control system **12** according to an embodiment of the present invention. The control system **12** adjusts the volume of a head-mounted display **22** in response to a gesture made by an occupant wearing the head-mounted display **22** among occupants riding in a vehicle **10**. It should be noted that, in the present specification, the occupant wearing the head-mounted display **22** is referred to as a user U.

[0025] The control system **12** includes a vehicle interior camera **14**, a vehicle exterior camera **16**, one or more vehicle interior microphones **18**, a vehicle exterior microphone **20**, and the head-mounted display **22**. The vehicle interior camera **14** and the head-mounted display **22** can perform wireless communication with each other. The vehicle exterior camera **16** and the head-mounted display **22** can perform wireless communication with each other. The plurality of vehicle interior microphones **18** and the head-mounted display **22** can perform wireless communication with each other. The vehicle exterior microphone **20** and the head-mounted display **22** can perform wireless communication with each other. It should be noted that the head-mounted display **22** can communicate with a server device **26** via a communication line **24** such as the Internet.

[0026] The vehicle interior camera **14** captures an image of the interior of the vehicle cabin of the vehicle **10**. The vehicle interior camera **14** transmits a vehicle interior image signal indicating the acquired vehicle interior image to the head-mounted display **22**. It should be noted that the vehicle interior camera **14** may be provided in plurality.

[0027] The vehicle exterior camera **16** captures an image of the surroundings of the vehicle **10**. The vehicle exterior camera **16** transmits a vehicle exterior image signal indicating the acquired vehicle exterior image to the head-mounted display **22**. It should be noted that the vehicle exterior camera **16** may be provided in plurality.

[0028] Each of the plurality of vehicle interior microphones **18** is disposed in the vehicle cabin of the vehicle **10**. For example, any one of the vehicle interior microphones **18** is disposed around each seat of the vehicle **10**. The vehicle interior microphones **18** each acquire a sound around the seat where the vehicle interior microphone **18** is disposed. Each of the vehicle interior microphones **18** transmits an acoustic signal indicating the acquired sound to the head-mounted display **22**.
[0029] The vehicle exterior microphone **20** is disposed outside the vehicle **10**. The vehicle exterior microphone **20** acquires a sound outside the vehicle. The vehicle exterior microphone **20** transmits an acoustic signal indicating the acquired sound to the head-mounted display **22**. It should be noted that the vehicle exterior microphone **20** may be provided in plurality.
[0030] FIG. **2** is a functional block diagram of the head-mounted display **22**. In the present specification, the head-mounted display **22** is also referred to as an HMD **22**. The HMD **22** includes an operation unit **30**, a display unit **32**, an acoustic unit **34**, a first communication unit **36**, a second communication unit **38**, a computation unit **40**, and a storage unit **42**.
[0031] The operation unit **30** is a human machine interface with which the user U can perform an input operation. The operation unit **30** may be constituted by a remote controller, a switch, or the like. The operation unit **30** outputs an operation signal corresponding to an operation performed by the user U, to the computation unit **40**.
[0032] The display unit **32** may be constituted by, for example, a display device. The display surface of the display device is close to both eyes of the user U. The display unit **32** displays (projects) a display image on the display surface based on an image signal transmitted from the computation unit **40**. The display unit **32** can provide the user U with a display image of the digital content, and provide the user U with a display image outside the vehicle. For example, the display unit **32** can display content related to augmented reality, mixed reality, or the like.
[0033] The acoustic unit **34** may be constituted by, for example, an audio device. The speaker (including a headphone and an earphone) of the acoustic unit **34** is close to an ear **62** (FIG. **4**A, FIG. **4**B, or the like) of the user U. The acoustic unit **34** outputs a sound from the speaker based on an acoustic signal transmitted from the computation unit **40**. The acoustic unit **34** can provide the user U with the sound of the digital content and provide the user U with the sound in the vehicle.
[0034] The first communication unit **36** may be constituted by, for example, a communication module (an integrated circuit module) including an antenna and the like. The first communication unit **36** receives signals transmitted from the vehicle interior camera **14**, the vehicle exterior camera **16**, the plurality of vehicle interior microphones **18**, and the vehicle exterior microphone **20**, and outputs the signals to the computation unit **40**.
[0035] The second communication unit **38** may be constituted by, for example, a communication module (an integrated circuit module) including an antenna and the like. The second communication unit **38** receives a digital content signal transmitted from the server device **26** via the communication line **24**, and outputs the digital content signal to the computation unit **40**.
[0036] The computation unit **40** may be constituted by a processor such as a central processing unit (CPU) or a graphics processing unit (GPU). That is, the computation unit **40** may be constituted by processing circuitry. At least part of the computation unit **40** may be realized by an integrated circuit such as an application specific integrated circuit (ASIC) or a field-programmable gate array (FPGA). At least part of the computation unit **40** may be realized by an electronic circuit including a discrete device.
[0037] The computation unit **40** includes a first acquisition unit **44**, a second acquisition unit **46**, a detection unit **48**, a display control unit **50**, and a volume control unit **52**. The first acquisition unit **44**, the second acquisition unit **46**, the detection unit **48**, the display control unit **50**, and the volume control unit **52** can be realized by the computation unit **40** executing a program stored in the storage unit **42**.
[0038] The first acquisition unit **44** acquires, via the first communication unit **36**, the vehicle interior image signal output from the vehicle interior camera **14**. Further, the first acquisition unit

44 acquires, via the first communication unit **36**, the vehicle exterior image signal output from the vehicle exterior camera **16**. In addition, the first acquisition unit **44** acquires, via the first communication unit **36**, the acoustic signal output from each of the vehicle interior microphones **18**. Further, the first acquisition unit **44** acquires, via the first communication unit **36**, the acoustic signal output from the vehicle exterior microphone **20**. On the other hand, the second acquisition unit **46** acquires, via the second communication unit **38**, the digital content signal output by the server device **26**.

[0039] The detection unit **48** detects a gesture performed by the user U, based on the vehicle interior image signal acquired by the first acquisition unit **44**. For example, the detection unit **48** detects the user U wearing the HMD **22** by image recognition. Further, the detection unit **48** detects the position of a hand **64** (FIG. **4**A, FIG. **4**B, or the like) of the user U and the movement of the fingers of the user U by gesture recognition.

[0040] The display control unit **50** causes the display unit **32** to display a display image corresponding to the vehicle exterior image signal acquired by the first acquisition unit **44**. Further, the display control unit **50** causes the display unit **32** to display a display image corresponding to the digital content signal acquired by the second acquisition unit **46**. The display control unit **50** can execute control of causing the display unit **32** to display the display image of the digital content and the display image of the vehicle exterior in a superimposed manner. The display control unit **50** can control the display unit **32** based on the operation signal output by the operation unit **30**.

[0041] The volume control unit **52** causes the acoustic unit **34** to output a sound corresponding to the acoustic signal acquired by the first acquisition unit **44**. Further, the volume control unit **52** causes the acoustic unit **34** to output a sound corresponding to the digital content signal acquired by the second acquisition unit **46**. The volume control unit **52** can execute control of causing the acoustic unit **34** to simultaneously output the sound of the digital content, the sound inside the vehicle cabin, and the sound outside the vehicle. The volume control unit **52** can execute control of adjusting the volume of the speaker of the acoustic unit **34** based on the gesture detected by the detection unit **48**. The volume control unit **52** can execute control of adjusting the volume of digital content based on the operation signal output by the operation unit **30**.

[0042] The storage unit **42** is a computer-readable storage medium. The storage unit **42** is constituted by a volatile memory (not shown) and a non-volatile memory (not shown). The volatile memory is, for example, a random access memory (RAM) or the like. The non-volatile memory is, for example, a read only memory (ROM), a flash memory, or the like. Data and the like are stored in, for example, the volatile memory. Programs, tables, maps, and the like are stored in, for example, the non-volatile memory. At least part of the storage unit **42** may be included in the processor, the integrated circuit, or the like described above.

2. Volume Adjustment Process

[0043] FIG. **3** is a flowchart of a volume adjustment process. The volume adjustment process described below is performed by the computation unit **40** of the HMD **22** at predetermined time intervals. The volume adjustment process described below is a process for adjusting the volume of the sound inside the vehicle cabin and the volume of the sound outside the vehicle, among the sounds output by the acoustic unit **34**.

[0044] In step S**1**, the first acquisition unit **44** acquires a vehicle interior image signal from the vehicle interior camera **14** via the first communication unit **36**. In step S**2**, the first acquisition unit **44** acquires acoustic signals from the plurality of vehicle interior microphones **18** and the vehicle exterior microphone **20** via the first communication unit **36**. In step S**3**, the detection unit **48** detects a gesture performed by the user U, based on the vehicle interior image signal acquired in step S**1**.

[0045] In step S**4**, the detection unit **48** determines whether or not a specific gesture is detected. Examples of the specific gesture that can be detected in the present embodiment will be described later. In a case where the specific gesture is detected (step S**4**: YES), the process proceeds to step

S**5**. On the other hand, in a case where the specific gesture is not detected (step S**4**: NO), the process returns to step S**1**.

[0046] In step S**5**, the volume control unit **52** adjusts one of the volume of the sound inside the vehicle or the volume of the sound outside the vehicle in accordance with the gesture detected by the detection unit **48**. The speaker of the acoustic unit **34** outputs a sound at a volume corresponding to the control of the volume control unit **52**.

3. Example of Specific Gesture and Volume Adjustment

[0047] In the present embodiment, in a case where the detection unit **48** detects a specific gesture performed by the user U, the volume control unit **52** can increase or decrease the volume of the sound inside or outside the vehicle output by the acoustic unit **34**.

3-1. First Volume Adjustment Example

[0048] In the first volume adjustment example, the volume control unit **52** increases or decreases the volume of the sound acquired by the plurality of vehicle interior microphones **18** and the volume of the sound acquired by the vehicle exterior microphone **20**, among the volumes output by the acoustic unit **34**.

[0049] As shown in FIGS. **4**A and **4**B, the user U can perform a gesture of positioning a palm **60** around the ear **62** without covering the ear **62** with the palm **60**. For example, as shown in FIG. **4**A, the user U can perform a first gesture of directing the palm **60** toward the front of the head and positioning the palm **60** behind the ear **62**. Further, as shown in FIG. **4**B, the user U can perform a second gesture of directing the palm **60** toward the rear of the head and positioning the palm **60** in front of the ear **62**. In a case where the first gesture or the second gesture is detected by the detection unit **48**, the volume control unit **52** increases the volume of the sound acquired by the plurality of vehicle interior microphones **18** and the volume of the sound acquired by the vehicle exterior microphone **20**, among the volumes output by the acoustic unit **34**.

[0050] As shown in FIG. **5**, the user U can perform a third gesture of covering the ear **62** with the hand **64**. In a case where the third gesture is detected by the detection unit **48**, the volume control unit **52** decreases the volume of the sound acquired by the plurality of vehicle interior microphones **18** and the volume of the sound acquired by the vehicle exterior microphone **20**, among the volumes output by the acoustic unit **34**.

3-2. Second Volume Adjustment Example

[0051] In the second volume adjustment example, the volume control unit **52** selectively increases or decreases the volume of the sound acquired by the plurality of vehicle interior microphones **18** and the volume of the sound acquired by the vehicle exterior microphone **20**, among the volumes output by the acoustic unit **34**.

[0052] As shown in FIG. **6**A, the user U can perform the first gesture with both hands **64***a* and **64***b*. Further, as shown in FIG. **6**B, the user U can perform the second gesture with the both hands **64***a* and **64***b*. In a case where the gesture shown in FIG. **6**A or the gesture shown in FIG. **6**B is detected by the detection unit **48**, the volume control unit **52** selectively increases the volume of the sound acquired by the vehicle exterior microphone **20**, among the volumes output by the acoustic unit **34**.

[0053] As shown in FIG. **7**, the user U can perform the first gesture or the second gesture with one hand **64***a* and lower the other hand **64***b*. In a case where the gesture shown in FIG. **7** is detected by the detection unit **48**, the volume control unit **52** selectively increases the volume of the sound acquired by the vehicle interior microphone **18** located in the direction in which the palm **60** of the one hand **64***a* is directed, among the volumes output by the acoustic unit **34**.

[0054] As shown in FIG. **8**, the user U can perform the first gesture or the second gesture with the one hand **64***a* and point in a direction with the other hand **64***b*. In a case where the gesture shown in FIG. **8** is detected by the detection unit **48**, the volume control unit **52** selectively increases the volume of the sound acquired by the vehicle interior microphone **18** located in the direction pointed by the other hand **64***b*, among the volumes output by the acoustic unit **34**.

[0055] As shown in FIG. **9**, the user U can perform the third gesture with the one hand **64***a* and point in a direction with the other hand **64***b*. In a case where the gesture shown in FIG. **9** is detected by the detection unit **48**, the volume control unit **52** selectively decreases the volume of the sound acquired by the vehicle interior microphone **18** located in the direction pointed by the other hand **64***b*, among the volumes output by the acoustic unit **34**.

4. Other Embodiment

[0056] The vehicle interior camera **14**, the vehicle exterior camera **16**, the vehicle interior microphones **18**, and the vehicle exterior microphone **20** may communicate with a communication device of the vehicle **10**. Further, the HMD **22** may communicate with the communication device of the vehicle **10**. That is, the devices (the vehicle interior camera **14**, the vehicle exterior camera **16**, the vehicle interior microphones **18**, and the vehicle exterior microphone **20**) provided in the vehicle **10** and the HMD **22** may communicate with each other via the communication device of the vehicle **10**.

5. Advantageous Effects

[0057] According to the present embodiment, the detection unit **48** detects a gesture performed by the user U, based on the image acquired by the vehicle interior camera **14** that constantly captures images of the interior of the vehicle cabin, and therefore can stably detect a gesture of the user U.

[0058] In a case where the volume of the head-mounted display **22** is adjusted by a device such as a remote controller, the sense of immersion of the user U may be impaired. However, according to the above configuration, since the volume of the head-mounted display **22** can be adjusted by the gesture of the user U, the sense of immersion of the user U is less likely to be impaired.

[0059] According to the above configuration, the head-mounted display **22** displays the display image corresponding to the image acquired by the vehicle exterior camera **16** that captures images of the periphery of the vehicle **10**, and therefore, it is possible to suppress motion sickness of the user U.

[0060] As described above, according to the present embodiment, the head-mounted display **22** can be used satisfactorily in the vehicle cabin.

6. Supplementary Notes

[0061] The following supplementary notes are further disclosed in relation to the above-described embodiments.

Supplementary Note 1

[0062] The control system (**12**) of the present disclosure includes: the head-mounted display (**22**) worn by the user (U) riding in the vehicle (**10**), the head-mounted display being configured to provide the user with at least a display image corresponding to an image acquired by the vehicle exterior camera (**16**) provided in the vehicle and configured to capture an image of the periphery of the vehicle; the detection unit (**48**) configured to detect a gesture performed by the user, based on an image acquired by the vehicle interior camera (**14**) provided in the vehicle and configured to capture an image of the interior of the vehicle cabin; and the volume control unit (**52**) configured to execute control of adjusting the volume of the speaker (**34**) provided in the head-mounted display based on the gesture detected by the detection unit.

[0063] According to the above configuration, the detection unit detects a gesture performed by the user, based on the image acquired by the vehicle interior camera that constantly captures images of the interior of the vehicle cabin, and therefore can stably detect a gesture of the user.

[0064] According to the configuration, since the volume of the head-mounted display can be adjusted based on the gesture performed by the user, the sense of immersion of the user is less likely to be impaired.

[0065] According to the above configuration, the head-mounted display displays the display image corresponding to the image acquired by the vehicle exterior camera that captures images of the periphery of the vehicle, and therefore, it is possible to suppress motion sickness of the user.

[0066] As described above, according to the above configuration, the head-mounted display can be used satisfactorily in the vehicle cabin.

Supplementary Note 2
[0067] In the control system according to Supplementary Note 1, the volume control unit may increase the volume in a case where the detection unit detects the gesture of positioning the palm (**60**) around the ear (**62**) without covering the ear with the palm.
Supplementary Note 3
[0068] In the control system according to Supplementary Note 1 or 2, the volume control unit may decrease the volume in a case where the detection unit detects the gesture of covering the ear with the hand (**64**).
Supplementary Note 4
[0069] In the control system according to any one of Supplementary Notes 1 to 3, in a case where the detection unit detects the gesture of positioning the palm of the left hand (**64***a*) around the left ear without covering the left ear with the palm of the left hand and positioning the palm of the right hand (**64***b*) around the right ear without covering the right ear with the palm of the right hand, the volume control unit may selectively increase the volume of the sound acquired by the microphone (**20**) configured to acquire the sound outside the vehicle.
Supplementary Note 5
[0070] In the control system according to any one of Supplementary Notes 1 to 4, in a case where the detection unit detects the gesture of positioning the palm of one hand (**64***a*) around the ear without covering the ear with the one hand and pointing in a direction with the other hand (**64***b*), the volume control unit may selectively increase the volume of the sound from the direction pointed by the other hand.
Supplementary Note 6
[0071] In the control system according to Supplementary Note 2, the volume of the sound from the direction in which the palm is directed may be selectively increased.
[0072] According to the configurations of Supplementary Notes 2 to 6, since the user can adjust the volume based on a gesture instead of an operation of the device, the sense of immersion of the user is less likely to be impaired.
Supplementary Note 7
[0073] The head-mounted display of the present disclosure is worn by the user riding in the vehicle, and includes: the display unit (**32**) configured to provide the user with at least a display image corresponding to an image acquired by the vehicle exterior camera provided in the vehicle and configured to capture an image of the periphery of the vehicle; the speaker configured to provide the user with a sound; the detection unit configured to detect a gesture performed by the user, based on an image acquired by the vehicle interior camera provided in the vehicle and configured to capture an image of the interior of the vehicle cabin; and the volume control unit configured to execute control of adjusting the volume of the speaker based on the gesture detected by the detection unit.
[0074] According to the above configuration, the detection unit detects a gesture performed by the user, based on the image acquired by the vehicle interior camera that constantly captures images of the interior of the vehicle cabin, and therefore can stably detect a gesture of the user.
[0075] According to the configuration, since the volume of the head-mounted display can be adjusted based on the gesture performed by the user, the sense of immersion of the user is less likely to be impaired.
[0076] According to the above configuration, the head-mounted display displays the display image corresponding to the image acquired by the vehicle exterior camera that captures images of the periphery of the vehicle, and therefore, it is possible to suppress motion sickness of the user.
[0077] Although the present disclosure has been described in detail, the present disclosure is not limited to the above-described individual embodiments. Various additions, replacements, modifications, partial deletions, and the like can be made to these embodiments without departing from the gist of the present disclosure, or without departing from the essence of the present

disclosure derived from the claims and equivalents thereof. Further, these embodiments can also be implemented in combination. For example, in the above-described embodiments, the order of operations and the order of processes are shown as examples, and are not limited to these. Furthermore, the same applies to a case where numerical values or mathematical expressions are used in the description of the above-described embodiments.

Claims

1. A control system comprising: a head-mounted display that is worn by a user riding in a vehicle, the head-mounted display being configured to provide the user with at least a display image corresponding to an image acquired by a vehicle exterior camera provided in the vehicle and configured to capture an image of a periphery of the vehicle; and one or more processors that execute computer-executable instructions stored in a memory, wherein the one or more processors execute the computer-executable instructions to cause the control system to: detect a gesture performed by the user, based on an image acquired by a vehicle interior camera provided in the vehicle and configured to capture an image of an interior of a vehicle cabin; and execute control of adjusting a volume of a speaker provided in the head-mounted display, based on the gesture that has been detected.
2. The control system according to claim 1, wherein the one or more processors cause the control system to increase the volume in a case where a gesture of positioning a palm around an ear without covering the ear with the palm is detected.
3. The control system according to claim 1, wherein the one or more processors cause the control system to decrease the volume in a case where a gesture of covering an ear with a hand is detected.
4. The control system according to claim 1, wherein the one or more processors cause the control system to selectively increase the volume of a sound acquired by a microphone configured to acquire a sound outside the vehicle, in a case where a gesture of positioning a palm of a left hand around a left ear without covering the left ear with the palm of the left hand and positioning a palm of a right hand around a right ear without covering the right ear with the palm of the right hand is detected.
5. The control system according to claim 1, wherein the one or more processors cause the control system to, in a case where a gesture of positioning a palm of one hand around an ear without covering the ear with the palm of the one hand and pointing in a direction with another hand is detected, selectively increase the volume of a sound from the direction pointed by the other hand.
6. The control system according to claim 2, wherein the one or more processors cause the control system to selectively increase the volume of a sound from a direction in which the palm is directed.
7. A head-mounted display that is worn by a user riding in a vehicle, the head-mounted display comprising: a display unit configured to provide the user with at least a display image corresponding to an image acquired by a vehicle exterior camera provided in the vehicle and configured to capture an image of a periphery of the vehicle; a speaker configured to provide the user with a sound; and one or more processors that execute computer-executable instructions stored in a memory, wherein the one or more processors execute the computer-executable instructions to cause the head-mounted display to: detect a gesture performed by the user, based on an image acquired by a vehicle interior camera provided in the vehicle and configured to capture an image of an interior of a vehicle cabin; and execute control of adjusting a volume of the speaker based on the gesture that has been detected.